

All-order compact-support modular density by one mollification

Run `fa.banach_001`, lane 1

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Status

Candidate full affirmative solution of Conjecture 1. The argument works for every integer $m \geq 1$ under either set of hypotheses printed in the source. It extracts the scalar content of the source's Lemma 12 and combines it with the ordinary $W^{m,1}$ approximate-identity theorem.

Source conjecture

The source is Y. Ahmida, I. Chlebicka, P. Gwiazda, and A. Youssfi, *Gossez's approximation theorems in the Musielak–Orlicz–Sobolev spaces*, arXiv:1711.06145. Conjecture 1 appears on PDF page 7. It defines

$$V_0^m L_M(\Omega) = \{u \in W_0^{1,1}(\Omega) \cap W^{m,1}(\Omega) : D^m u \in L_M(\Omega)\}$$

and asks whether every u in this space with compact support in Ω has $u_k \in C_c^\infty(\Omega)$ such that all derivatives through order m converge in L^1 , while $D^m u_k$ converges modularly in L_M .

1) $C_0^\infty(\overline{\Omega})$ is $\sigma(\Pi L_M, \Pi L_{M^*})$ -dense in $W^m L_M(\Omega)$.

2) $C_0^\infty(\Omega)$ is $\sigma(\Pi L_M, \Pi L_{M^*})$ -dense in $W_0^m L_M(\Omega)$ provided that M^* satisfies the hypothesis (M1).

The Musielak–Orlicz spaces and PDEs

It can be useful in analysis of PDEs (see e.g. [18, 20]) to provide a modular density result not for $W_0^m L_M(\Omega)$, but for

$$V_0^m L_M(\Omega) = \{u \in W_0^{1,1}(\Omega) \cap W^{m,1}(\Omega) : D^m u \in L_M(\Omega)\},$$

where $W_0^{1,1}(\Omega)$ is defined as a closure of $C_0^\infty(\Omega)$ in $W^{1,1}(\Omega)$ -norm. In general, $W_0^m L_M(\Omega) \neq V_0^m L_M(\Omega)$. Thus, we state below the conjecture. We solve the problem in Appendix for $m = 1$ only.

Conjecture 1. Assume that $\Omega \subset \mathbb{R}^N$ satisfies the segment property and an N -function M satisfies (M1) and (M2) (resp. (3), (M1), and (M2p)). Then for every $u \in V_0^m L_M(\Omega)$ with $\text{supp } u \subset\subset \Omega$, there exist $\lambda > 0$ and a sequence of functions $u_k \in C_0^\infty(\Omega)$ such that $D^\alpha u_k \xrightarrow[k \rightarrow \infty]{} D^\alpha u$ in $L^1(\Omega)$ for any $|\alpha| \leq m$ and $D^m u_k \xrightarrow[k \rightarrow \infty]{M} D^m u$ modularly in $L_M(\Omega)$.

Furthermore, in some applications if a construction of a function to approximate is known, another approach

Write

$$\rho_M(f) = \int_{\Omega} M(x, |f(x)|) dx.$$

Modular convergence means that $\rho_M((f_k - f)/\lambda) \rightarrow 0$ for some $\lambda > 0$.

The scalar observation hidden in Lemma 12

Scalar mollification lemma. *Assume the source's conditions $(\mathcal{M}1)$ and $(\mathcal{M}2)$; alternatively assume (3), $(\mathcal{M}1)$, and $(\mathcal{M}2p)$. If $F \in L_M(\Omega)$ has compact support in Ω , then*

$$J_\varepsilon * F \longrightarrow F \quad \text{modularly in } L_M(\Omega).$$

In the alternative power-growth case, use also the resulting $F \in L^p$.

Proof. This is exactly the single-field argument in the proof of source Lemma 12, equations (33)–(37). By source Lemma 4, choose bounded compactly supported F_n and one $\lambda > 0$ such that $\rho_M((F_n - F)/\lambda) \rightarrow 0$. Convexity gives

$$\begin{aligned} \rho_M\left(\frac{J_\varepsilon * F - F}{3\lambda}\right) &\leq \frac{1}{3}\rho_M\left(\frac{J_\varepsilon * (F - F_n)}{\lambda}\right) + \frac{1}{3}\rho_M\left(\frac{J_\varepsilon * F_n - F_n}{\lambda}\right) \\ &\quad + \frac{1}{3}\rho_M\left(\frac{F_n - F}{\lambda}\right). \end{aligned}$$

The source's mollifier estimate (Lemma 9) bounds the first term by a constant times the third, uniformly for small ε ; uniformity is precisely the limit condition in $(\mathcal{M}2)$ (respectively $(\mathcal{M}2p)$). For fixed n , source Lemma 5 gives translation continuity of F_n , and Jensen–Fubini therefore sends the middle term to zero. First choose n and then ε . Since the modular error of F_n is arbitrary, the claim follows. The source proof explicitly notes that the approximants need not be derivatives, so no Sobolev structure of F is used. $\square$

The source states Lemma 12 for $u \in W^m L_M$:

$$f\left(\sum_{i=1}^k \frac{a_i}{2^i}\right) = f\left(\sum_{i=1}^k \frac{2^{k-i}}{2^k-1} \frac{2^k-1}{2^{k-i}} \frac{a_i}{2^i}\right) \leq \sum_{i=1}^k \frac{2^{k-i}}{2^k-1} f\left(\frac{2^k-1}{2^{k-i}} \frac{a_i}{2^i}\right) \leq \sum_{i=1}^k \frac{2^{k-i}}{2^k-1} f(a_i).$$

In the following lemma we prove the convergence of the mollification. Such result in the case of Orlicz spaces can be found in [2, Lemma 3.16].

Lemma 12. *Assume that an N -function M satisfies $(\mathcal{M}1)$ and $(\mathcal{M}2)$ (resp. (3), $(\mathcal{M}1)$ and $(\mathcal{M}2p)$). If $u \in W^m L_M(\Omega)$ has a compact support in Ω and u_ε stands for the sequence defined in (8), then $u_\varepsilon \xrightarrow[\varepsilon \rightarrow 0^+]{\text{mod}} u$ in $W^m L_M(\Omega)$. If additionally $u \in W^m E_M(\Omega)$, then $u_\varepsilon \rightarrow u$ in norm in $W^m E_M(\Omega)$.*

Its proof applies the preceding scalar argument separately to each $D^\alpha u$. Retaining only one field gives the lemma above.

Full proof of Conjecture 1

Theorem. *Conjecture 1 holds for every $m \geq 1$.*

Proof. Let $u \in V_0^m L_M(\Omega)$ and let $K = \text{supp } u \Subset \Omega$. Extend u by zero to $\mathbb{R}^N$. Because u vanishes on a neighborhood of $\partial\Omega$, this zero extension belongs to $W^{m,1}(\mathbb{R}^N)$ and its weak derivatives are the zero extensions of the derivatives on Ω .

Set $u_\varepsilon = J_\varepsilon * u$. If $0 < \varepsilon < \text{dist}(K, \partial\Omega)$, then $u_\varepsilon \in C_c^\infty(\Omega)$ and, for every multi-index $|\alpha| \leq m$,

$$D^\alpha u_\varepsilon = J_\varepsilon * D^\alpha u.$$

Since $D^\alpha u \in L^1(\mathbb{R}^N)$, the standard L^1 approximate-identity theorem yields

$$\|D^\alpha u_\varepsilon - D^\alpha u\|_{L^1(\Omega)} \longrightarrow 0 \quad (|\alpha| \leq m).$$

For $|\alpha| = m$, the defining condition of $V_0^m L_M$ gives $D^\alpha u \in L_M(\Omega)$, and its support lies in K . The scalar mollification lemma therefore gives

$$J_\varepsilon * D^\alpha u \longrightarrow D^\alpha u \quad \text{modularly in } L_M(\Omega).$$

There are only finitely many order- m components. If their number is q and their modular scales are λ_α , choose $\Lambda = q \max_\alpha \lambda_\alpha$ and use $|D^m u_\varepsilon - D^m u| \leq \sum_{|\alpha|=m} |D^\alpha u_\varepsilon - D^\alpha u|$ together with convexity of M ; this combines the componentwise statements into modular convergence of $D^m u_\varepsilon$.

Taking any sequence $\varepsilon_k \downarrow 0$ below the support distance and putting $u_k = u_{\varepsilon_k}$ proves every conclusion of Conjecture 1. $\square$

Notice that the segment property is unnecessary for this compact-interior case; it is used elsewhere in the source to move supports that meet the boundary.

References

- [1] Y. Ahmida, I. Chlebicka, P. Gwiazda, and A. Youssfi, *Gossez's approximation theorems in the Musielak–Orlicz–Sobolev spaces*, J. Funct. Anal. 275 (2018), 1691–1714; arXiv:1711.06145.